Shubham Kaushik

shubhamkaushik.com | LinkedIn | GitHub | kaushiks[at]brandeis[dot]edu

Systems engineer focused on storage engine design, file system internals, and distributed systems using C/C++. Published at VLDB, ICDE, TPCTC on scheduling, caching, and metadata management to optimize data systems.

EDUCATION

Jan 2024 - Present	**Doctor of Philosophy** in *Computer Science* (GPA: 3.97/4.0) Brandeis University, MA, United States
Sep 2022 - Dec 2023	**Master of Science** in *Computer Science "Data-Centric Computing"* (GPA: 3.88/4.0) Boston University, MA, United States
Jul 2014 - Jun 2018	**Bachelor of Technology** in *Computer Science & Engineering* Maharshi Dayanand University, India

EXPERIENCE

Jan 2024 - Present	**Ph.D. Researcher** @ Smart & Scalable Data Systems Lab, Brandeis University
Mar 2022 - Aug 2022	**Software Engineer** @ Server Programming Team, Kwalee, India
Jun 2021 - Mar 2022	**Engineer - Information Security** @ Cyber Fusion, FIS Global, India
Jul 2018 - Jun 2021	**Project Engineer** @ Cloud Computing - Wipro Digital, Wipro Limited, India

TECHNICAL SKILLS

- **Languages & Tools**: C/C++, Python, Rust, RAG, LLM
- **Markup Languages**: HTML, CSS, JSON, YAML, LaTeX, Markdown
- **Storage & Databases**: RocksDB, LSM-trees, SQL, SQLite, ORM
- **Tools & Systems**: Linux, Asyncio, Git, Docker, Django, Django-REST

KEY PROJECTS

Tectonic [Paper PDF] [GitHub] **Apr 2025 - Present**

- Designed and implemented Tectonic in Rust, a high-performance workload generator for key-value storage systems, achieving 2× higher throughput through lock-free data structures and zero-copy I/O patterns.
- Reduced memory footprint by 84% via precise memory management and buffer pool optimization, enabling accurate storage system benchmarking under resource constraints.

LSM Buffer Analysis [Paper PDF] [Blog] [GitHub] **Jan 2024 - Present**

- Profiled and benchmarked in-memory buffer implementations (skiplist, vector, hash-hybrids) for LSM-based storage engines; analyzed concurrency, caching, and eviction behavior across workloads for storage tuning.
- Built a workload characterization framework mapping I/O access patterns to optimal buffer configurations, enabling systematic storage engine design decisions.

RangeReduce [Paper PDF] [Blog] [GitHub] **Sep 2023 - Feb 2026**

- Developed RangeReduce, a query-driven compaction engine that adapts file system-level data layout to live range query patterns, reducing disk I/O amplification and CPU overhead in production LSM storage systems.
- Achieved 18% lower range-query latency, 90% reduction in compaction debt, 12% less data movement, and 20% lower space utilization — validated through structured benchmarking across diverse workload profiles.

SELECTED PUBLICATIONS

VLDB 2026	Abhishek Chanda*, **Shubham Kaushik***, Artem Lavrov, and Subhadeep Sarkar. *TexBench: A Unified Benchmarking Suite for Shifting Workloads*, 52nd International Conference on Very Large Data Bases
ICDE 2026	**Shubham Kaushik**, Manos Athanassoulis, and Subhadeep Sarkar. *RangeReduce: Query-Driven Compaction In LSM-Trees*, 42nd IEEE International Conference on Data Engineering [Talk]
TPCTC 2025	Alexander H. Ott*, **Shubham Kaushik***, James Chen, and Subhadeep Sarkar. *Tectonic: Bridging Synthetic and Real-World Workloads for Key-Value Benchmarking*, TPC Technology Conference on Performance Evaluation & Benchmarking
DBTest 2024	**Shubham Kaushik** and Subhadeep Sarkar. *Anatomy of the LSM Memory Buffer: Insights & Implications*, In Proceedings of the International Workshop on Testing Database Systems

CURRICULAR ACTIVITIES

- Member, VLDB 2026 Shadow PC
- Judged and mentored at *HackMIT 2023* and *BostonHacks 2022*